

A Decoder-Only Jacobian Lens for Speech-to-Text Models

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Abstract—The Jacobian lens identifies representations a language model is poised to verbalize by averaging causal maps from a layer’s residual stream to the final pre-logit state. Encoder-decoder speech recognizers split that residual across two stacks: an encoder that is never unembedded and a decoder that is. We show that a pooled encoder-to-decoder transport is not this object—it destroys the causal (t, t') structure and leaves the unembedding off-manifold. We instead fit one Jacobian per decoder block, $J_\ell = \mathbb{E}_{t, t' \geq t} [\partial h_{\text{final}, t'} / \partial h_{\ell, t}]$, with Hutchinson probes, and read out $\text{softmax}(E J_\ell h_{\ell, t})$ through Whisper’s own tied projection. The construction is implemented in ECHO. On a linear stub decoder the estimator recovers the closed form $J = 2M/(T + 1)$ to relative Frobenius error < 0.1 ; a Whisper-base fit yields six 512×512 maps. The resulting (position $\times$ layer) grid is a vocabulary-space audit of what the decoder is poised to say: it supports substitution and hallucination inspection, prior-versus-evidence maps, and choosing which decoder depth to probe—questions that attention and saliency do not answer. A workspace-band census is left as future work.

Index Terms—speech recognition, mechanistic interpretability, Jacobian lens, Whisper, encoder-decoder Transformers

I. INTRODUCTION

Whisper [1] and related encoder-decoder ASR models transcribe by unembedding a decoder residual stream that has attended to a frozen acoustic encoder. Failures such as silence-induced repetition, homophone substitution, and early cutoff are well known, yet the decoder state that is about to become the next token is rarely inspected in vocabulary space.

Gurnee et al. [2] give language models a tool for that inspection: the Jacobian lens averages $\partial h_{\text{final}, t'} / \partial h_{\ell, t}$ over causally allowed pairs $t' \geq t$, then reads the transported residual with the model’s unembedding. The ranked tokens are what the model is *poised to verbalize*. The construction assumes a single residual stream that ends in an unembedding. Speech-to-text models do not: the encoder never sees that unembedding.

This paper ports the estimator to the stream that does. We fit decoder-only Jacobians for seq2seq ASR, estimate them with Hutchinson probes [3] so the encoder can sit outside the autograd graph, and ship fit/apply as jobs in ECHO, an audio interpretability workbench in the spirit of LIT [4]. We also record why a pooled encoder transport—the obvious first attempt—is a different, weaker object. Practically, the tool is for people who already have a transcript (or a failure) and need to know whether the decoder was ranking the spoken word, a language-model collocation, a silence loop, or an early stop—without claiming that Whisper has a global workspace. The

claim is narrower: if one wants the LLM lens, one must fit it on the decoder.

II. RELATED WORK

The logit lens [5] applies the unembedding to intermediate residuals (identity transport). The tuned lens [6] learns an affine map to the final layer. The Jacobian lens [2] is the average *derivative* of the true residual-to-final map. We copy that estimator, not the LLM write/hold/ablate suite.

DecoderLens [7] feeds intermediate *encoder* layers to a frozen decoder’s cross-attention so the decoder can verbalize encoder states. Langedijk et al. already run this on Whisper transcription: early encoder layers emit empty or repetitive strings; later ones approximate the transcript. Complementary: they interpret the encoder *through* the decoder; we interpret the decoder *with* its own Jacobian. Linear probes [8]–[10] test linearly available features but are not maps onto the model’s unembedding.

ECHO is the audio analogue of LIT [4]: attention, saliency, embeddings, and perturbations on Whisper and wav2vec 2.0 [11]. Attention says where the decoder looked in the audio; saliency says which waveform regions moved a scalar. Neither says what the decoder is poised to say. CTC models have no decoder residual to unembed and are excluded.

III. METHOD

A. Substrate

Whisper is an encoder-decoder Transformer [12]. A 30 s log-Mel spectrogram (80 bins $\times$ 3000 frames on the base model) is downsampled by a stride-2 convolution to $n_{\text{audio}} = 1500$ encoder positions of width $d = 512$. The decoder, at each transcript position $t \leq 448$, attends causally to its prefix and, via cross-attention, to all 1500 encoder frames, then unembeds with a tied projection $E \in \mathbb{R}^{V \times d}$ ($V = 51865$). We write $h_{\ell, t} \in \mathbb{R}^d$ for the residual at decoder block ℓ and position t , and $h_{\text{final}, t'}$ for the post-final-LayerNorm state that E consumes.

Current HuggingFace Whisper records decoder hidden states from each decoder block only. The embedding output is not in that tuple; h_{final} is a distinct tensor (final LayerNorm of the last block). The lens therefore has *one map per decoder block*: six on Whisper-base, matching a fitted artifact on this codebase. The encoder is computed once without gradients and is never differentiated.

B. Estimator

Following [2], the lens at block ℓ is

$$J_\ell = \mathbb{E}_{t, t' \geq t, \text{clip}} \left[\frac{\partial h_{\text{final}, t'}}{\partial h_{\ell, t}} \right] \in \mathbb{R}^{d \times d}. \quad (1)$$

The model is frozen. There is no optimizer and no learned baseline. Fit is teacher-forced: labels are shifted right past the decoder-start token so all T positions run in one pass.

Exact Jacobians are unnecessary. For a Rademacher probe $r \in \{\pm 1\}^d$, the scalar $\sum_{t'} \langle h_{\text{final}, t'}, r \rangle$ has vector-Jacobian product $g_t = \sum_{t' \geq t} J_\ell(t \rightarrow t')^\top r$ at every source t at once, because the causal mask zeroes $t > t'$. The identity $\mathbb{E}[rr^\top] = I$ then gives

$$\hat{J}_\ell = \frac{1}{NP \cdot T(T+1)/2} \sum_{n=1}^N \sum_{p=1}^P r^{(p)} \left(\sum_t g_t^{(n,p)} \right)^\top, \quad (2)$$

with N clips and P probes [3]. Product defaults are $P = 4$ (range 1–32) and $N \in [2, 1000]$. Audio is truncated at 30 s (cap 60 s); Whisper still pads Mel to 30 s.

C. Readout

Apply re-runs the decoder. Positions come from the model’s generation routine (Whisper’s decoding mix-in) or from a provided transcript. A teacher-forced pass collects $h_{\ell, t}$. Each cell is

$$\text{lens}(h_{\ell, t}) = \text{softmax}(E J_\ell h_{\ell, t}), \quad (3)$$

implemented as $h J_\ell^\top E^\top$. Whisper’s final LayerNorm is already inside h_{final} , so it is absorbed into J_ℓ ; we do not apply an extra RMS-norm (unlike [2]). Top- k (default $k = 5$) ranks logits; the softmax is for display, not calibrated confidence. Neighboring cells are not a phrase.

The readout axis is transcript position, not audio time. Which encoder frame supports a token remains a cross-attention question.

D. Why encoder transports fail

An earlier design fitted a global map from the mean-pooled encoder state to the mean-pooled decoder state and applied it to 0.3 s encoder buckets (Table I). Differentiating after pooling erases (t, t') alignment; the linearization is around clip means but is applied locally; the unembedding is used off-manifold with the constant dropped. That object is not (1). CTC support came for free (the encoder is the only stream) and disappeared with the redesign: a CTC head never verbalizes through a decoder workspace.

IV. IMPLEMENTATION AND VERIFICATION

ECHO stores a format-v2 artifact: architecture *decoder*, method Hutchinson-decoder-VJP, one square $d \times d$ matrix per recorded decoder block, bound to a model revision. Apply refuses a mismatch. Only seq2seq ASR adapters expose the architecture; wav2vec 2.0 returns none. Fig. 1 sketches the data flow. A production fit on this codebase uses Whisper-base, $N = 1000$ transcripts, $P = 4$, and writes six 512×512 maps.

TABLE I
POOLED ENCODER TRANSPORT VERSUS THE DECODER-ONLY LENS.

	Encoder transport	Decoder-only J-lens
Source	mean of 1500 encoder frames	residual $h_{\ell, t}$
Target	mean of T decoder positions	$h_{\text{final}, t'}, t' \geq t$
Spaces	encoder $\rightarrow$ decoder	decoder $\rightarrow$ decoder
Readout	$(h - \bar{h}) J^\top E^\top$	$\text{softmax}(E J h)$
Units	0.3 s buckets	decoder tokens
CTC	yes	no

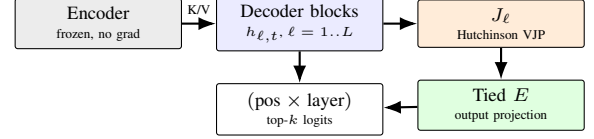


Fig. 1. Decoder-only J-lens in ECHO. The encoder supplies cross-attention keys and values only. Each decoder block has its own J_ℓ ; readout uses the model’s tied E .

a) *Linear closed form.*: A unit-test stub uses a position-wise linear decoder $h_{\text{final}, t} = M h_{\ell, t}$ with no mixing across positions. Then $\partial h_{\text{final}, t'} / \partial h_{\ell, t}$ is M on the diagonal $t' = t$ and zero otherwise, so the triangular mean in (1) is exactly $2M/(T+1)$. With $P = 512$ Rademacher probes the fitted map matches this matrix to relative Frobenius error below 0.1. Apply on the same stub matches an independent recomputation of $(Jh)E^\top$ at a held-out cell.

These tests check the estimator and the readout identity. They are not a Whisper workspace census.

V. WHAT THE ANALYSIS PROVIDES

A fitted lens is reused. Each apply job on one clip returns a grid (Fig. 2): rows are decoder blocks (UI “Layer $\ell+1$ ” for 0-based ℓ), columns are decoder input tokens (generated, or a provided reference), and each cell is a top- k ranking with logit scores. That ranking is the analysis. It is not a second transcript, not an attention map, and not a confidence calibrated to word error.

Three properties distinguish it from the other ECHO tools. Attention answers *where* a decoder position looked in the 1500 encoder frames. Saliency answers *which waveform region* moved a scalar score. The J-lens answers *what token the decoder residual is aligned with* at that depth, in the model’s own vocabulary, after replacing everything downstream of the block by the average causal map (1). The three views are meant to be used together: attention for alignment, saliency for the waveform, the grid for verbal content.

The readings below are *affordances of that contract*—what an analyst can ask the grid—not measured effects on a corpus. A cell is a first-order, context-averaged disposition, not a guaranteed continuation.

A. How to read one clip

Start-of-transcript. The first decoder input is the start token. No transcribed word is in the prefix. The residual still

contains that token’s embedding and position; audio enters only through cross-attention. The cell is the closest available reading of what the audio wrote into the decoder before any transcript language-model context: likely first words, Whisper’s silence convention, or background-speech artifacts.

Depth trajectory (a column). Read one position from early blocks (often frequent-token geometry) toward later blocks (closer to the emitted word). Where a content word first enters the top- k is a candidate crystallization depth—the depth at which acoustic evidence is integrated into a verbal decision. A column that never ranks the emitted word is a candidate for transcription against the decoder’s internal ranking: the model said the word, but the residual was not poised to say it.

Prior versus evidence (mid vs. late at the same t). Agreement of mid and late blocks on a collocation is consistent with a language-model prior; the audio could have been almost anything. A late-block flip to an acoustically specific or rare word is consistent with audio-driven selection. Sweeping that disagreement along the transcript is a per-word prior-versus-listening map. It is a comparison of rankings, not a causal mediation test.

Ambiguity, loops, endpoints. A wide top- k (their/there/they’re, or two number names) is consistent with under-determined acoustics and a fallback to prior; a narrow gap is consistent with the acoustics having decided. Rankings of a repetition token *before* generation emits it would be a leading indicator of Whisper’s silence loop. Drift toward end-of-text or trained closers (“thank you”) while speech continues would be an early-cutoff signature; its absence on run-on audio is the complementary runaway-transcription signature.

B. Practical use

Table II maps analyst questions onto those readings. The intended user is someone already looking at a clip in ECHO—a researcher debugging Whisper, a practitioner auditing a high-stakes transcript, or a student asking why two models diverged.

Substitution audit. When the hypothesis is “the model heard A but wrote B ,” apply with the generated transcript and read the column at B . If A never enters the top- k , the decoder was not poised to say the spoken word (a listening failure, or an encoder/cross-attention failure—check attention next). If A ranks in mid blocks and B only at the top, the residual had the spoken word and the unembedding path preferred a collocation. That distinction is the difference between collecting more audio and changing decoding or prompting.

Hallucination and silence loops. Whisper’s familiar failure under padded silence is a repetition loop. Apply on the failing clip and scan cells *before* the first repeated token. Loop tokens ranked highly in mid blocks while the input token is unrelated are a precursor in vocabulary space: the direction is visible before it is spoken. The same procedure flags confabulated content words under noise. This is a diagnostic for a single clip, not a detector with a reported false-positive rate.

Ambiguity and high-stakes tokens. For names, numbers, and homophones, sort positions by top- k score gap. Wide cells

TABLE II
ANALYST QUESTIONS, WHAT TO READ, AND THE COMPLEMENTARY ECHO TOOL.

Question	Read	Complements
Spoken word ranked?	column at the error	cross-attention
Prior or audio?	mid vs. late, same t	—
Loop forming?	cells before the repeat	waveform / silence
Ambiguous token?	top- k score gap	—
Noise changed it?	same column, clean vs. noisy	perturbation WER
Which depth to probe?	row where top-1 first matches	layer probes

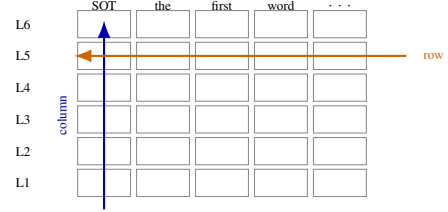


Fig. 2. Schematic (position $\times$ layer) grid. A *column* is a depth trajectory at one token; a *row* is one block along the transcript. Cells are ranked verbalizable tokens, not a generated phrase.

are where a human reviewer should look first. The lens does not replace a language model confidence score; it shows the *competitors* in the decoder residual, which a scalar WER or log-probability does not.

Clean versus perturbed audio. ECHO already applies noise, masks, and pitch shift. Applying the same lens to the clean clip and the perturbed clip, with the same reference transcript, shows whether crystallization depth moved, whether a late-block ranking flipped, or whether a loop precursor appeared—a decoder-state view of robustness complementary to WER.

Generated versus teacher-forced. Apply twice: once with generated positions and once with a gold transcript. Disagreement in a column is a candidate for free-running dynamics that teacher-forcing hides. Fit itself is teacher-forced, so this comparison is an inspection, not a claim that J models the sampler.

Where to probe. Layer probes and interventions are expensive if run on every block. A cheap apply pass suggests which rows are worth reading (where top-1 first matches the next token) and which can be ignored as frequent-token noise. Confirming that band with a corpus-level top-1 agreement curve—the speech analogue of the LLM workspace band [2]—is not reported here.

What the analysis is *not*: it does not name the supporting audio frame; it does not give calibrated word confidence; it does not steer or ablate the model; it does not apply to CTC. Those jobs stay with cross-attention, proper uncertainty estimators, a future write mode, and encoder-side tools respectively.

VI. LIMITATIONS

J_ℓ is a first-order average: trust rankings more than scores, and expect error far from the fit distribution. Averaging over pairs erases which audio frame a token reads. Fit is teacher-forced, so J reflects “given this prefix,” not free-running dynamics. $P = 4$ leaves probe noise; the closed-form test used $P = 512$. Frequent tokens dominate geometry. The artifact is revision-bound. Write-mode interventions from [2] are unimplemented. Layer count follows whatever HuggingFace records as decoder hidden states; it is six on current Whisper-base, not an embedding-plus-blocks stack.

VII. CONCLUSION

The Jacobian lens is a statement about a residual that is about to be unembedded. In encoder–decoder ASR that residual is the decoder’s. Fitted there, with causal averaging and the model’s own E , it is the same estimator as in [2] and is already a job in ECHO: an analyst can audit a substitution, inspect a silence loop, or decide which decoder depth to probe, using the same vocabulary the model speaks. Fitted on pooled encoder states, it is not a lens. The measurements that would make the speech case as strong as the LLM case—workspace-band statistics, multi-size Whisper, and a write mode—should be taken on this decoder construction.

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